Zoology Legends

Z00501

Final Term Papers 2025

Muhammad Zaman

0304-4756496

Join our Legend Group

https://chat.whatsapp.com/CZOgFCOYuWMHF75gTJNMFg

Team Zoology LEGENDS

Protostomes

- From the Greek term meaning "mouth first."
- During embryonic development, the **mouth forms first** from the blastopore, and the anus develops later.
- Includes phyla like **mollusks, arthropods, and annelids (worms)**.

There are two major branches of the protostomes.

1) The **Ecdysozoa** includes those animals that molt. Its major constituent is Arthropoda, a phylum containing insects, arachnids, mites, crustaceans, and millipedes.

2) The second major group of protostomes are the **Lophotrochozoa**. They are characterized by a common type of cleavage (spiral), a common larval form, and a distinctive feeding apparatus.

These phyla include annelids, molluscs, and flatworms.

Deuterostomes

- From the Greek term meaning "mouth second."
- In this group, the **anus forms first** from the blastopore, followed by the mouth.
- Includes phyla like **chordates** (e.g., vertebrates) and **echinoderms** (e.g., starfish).

Three basic modes of commitment

The three basic modes of commitment in embryonic development are:

1. **Autonomous Specification**:

Cells acquire specific fates based on the differential distribution of cytoplasmic molecules in the egg.

- Characteristic of most invertebrates, leading to "mosaic" development where cells cannot change their fate if removed.
- In autonomous specification, certain cytoplasmic molecules present in the egg determine the fate of cells.
- This type of specification is common in most invertebrates.
- Cell type specification happens before any large-scale embryonic cell migration.
- This autonomous specification leads to a pattern of development called mosaic development. It’s like the embryo is made up of independent self-differentiating parts, just like a mosaic.
- Invariant cleavages occur, producing the same lineages in each embryo of the species.
- The fate of blastomeres are invariant.
- Produces “mosaic” (“determinative”) development: cells cannot change fate if a blastomere is lost.

2. **Conditional Specification**:

- Cells' fates depend on interactions with neighboring cells, making this mode regulative.
- Typical of vertebrates; cells can alter their fate if others are removed, allowing for compensation (e.g., identical twins).
- Conditional specification is a characteristic found in all vertebrates and a few invertebrates.
- o In this type of specification, cells interact with each other to determine their fate.
- o The relative positions of cells become important in this process.
- o Unlike in autonomous specification, the fate of cells is not predetermined and can vary.
- o Variable cleavages occur, meaning that cells do not have fixed assignments.
- Additionally, there are significant cell rearrangements and migrations that happen before or during specification.
- o Capacity for "regulative" development: allows cells to acquire different functions.

3. **Syncytial Specification**:

- Found in many insects, where interactions occur between different parts of a single multinucleate cell (syncytium).
- Insects like Drosophila use mRNA gradients (e.g., Bicoid and Nanos proteins) to specify body regions.

- Variable cleavage produces no rigid cell fates for particular nuclei

Two development stages of organisms

I. CLEAVAGE or Blastomere:

Cleavage is a series of extremely rapid mitotic divisions wherein the enormous volume of zygote cytoplasm is divided into numerous smaller cells. These cells are called **blastomeres**, and by the end of cleavage, they generally form a sphere known as a blastula.

II. GASTRULATION or Germ Layers:

A series of extensive cell rearrangements is called gastrulation, and the embryo is said to be in the gastrula stage. As a result of gastrulation, the embryo contains three **germ layers**: the ectoderm, the endoderm, and the mesoderm.

ʊ The **ectoderm** gives rise to the epidermis, nervous system, and pigment cells.

ʊ The **mesoderm** generates the kidneys, gonads, bones, heart, and blood cells.

ʊ The **endoderm** forms the lining of the digestive tube and the respiratory system.

III. ORGANOGENESIS:

Once the three germ layers are established, the cells interact with one another and rearrange themselves to produce tissues and organs. This process is called organogenesis. Many organs contain cells from more than one germ layer.

For example, the outer layer of skin comes from the ectoderm, while the inner layer (the dermis) comes from the mesoderm. Also during organogenesis, certain cells undergo long migrations from their place of origin to their final location.

Protostomes & It's Types

- From the Greek term meaning "mouth first."
- During embryonic development, the **mouth forms first** from the blastopore, and the anus develops later.
- Includes phyla like **mollusks, arthropods, and annelids (worms)**.

There are two major branches of the protostomes.

- The **Ecdysozoa** includes those animals that molt. Its major constituent is Arthropoda, a phylum containing insects, arachnids, mites, crustaceans, and millipedes.
- The second major group of protostomes are the **Lophotrochozoa**. They are characterized by a common type of cleavage (spiral), a common larval form, and a distinctive feeding apparatus. These phyla include annelids, molluscs, and flatworms.

Levels / Regulation of Gene expression

The regulation of gene expression controls when and how genes are expressed, and it can occur at multiple levels:

1. **Differential gene transcription**: This regulates which nuclear genes are transcribed into RNA. Not all genes are transcribed at the same time or at the same rate, depending on the cell's needs.
2. **Selective nuclear RNA processing**: After transcription, some RNAs may be processed differently, influencing which parts of the RNA are used to form messenger RNA (mRNA) that will leave the nucleus.
3. **Selective mRNA translation**: In the cytoplasm, some mRNAs are selected for translation into proteins, while others are not, ensuring only necessary proteins are produced.
4. **Differential protein modification**: After proteins are made, they can be modified (such as through phosphorylation) to regulate their function or stability in the cell.

Transcription factors domains

Transcription factors have three major domains.

1. **DNA-binding domain**
2. **Protein-protein interaction domain**
3. **Trans-activating domain**

➢ **DNA-binding domain**:

It recognizes a particular DNA sequence.

➢ **Protein-protein interaction domain**:

It allows the transcription factor's activity to be modulated by TAFs or other transcription factors.

➢ **Trans-activating domain**:

This domain activates or represses transcription by interacting with RNA polymerase and other components of the transcription machinery. It can either promote or inhibit the transcription of genes by modifying the transcriptional process.

Parts of egg (Anatomy of an Egg)

An egg consists of four main parts:

1. **Shell**:
    - Outer covering made of calcium carbonate.
    - Has ~17,000 pores, allowing air and moisture to pass.
    - Provides protection against bacteria.
2. **Shell Membranes**:
    - Two transparent layers inside the shell (Inner and outer Shell Membrane).
    - Made of keratin, offering additional defense against bacteria.
3. **Albumen (Egg White)**:
    - Makes up 67% of the egg's liquid weight.
    - Contains over half the egg's protein.

4. **Yolk**:
    - Nutrient-rich part with fats, vitamins, and minerals.
    - Color varies based on the hen's feed and breed.
    - These include iron, vitamin A, vitamin D, phosphorus, calcium, thiamine, and riboflavin.

Difference between Karyokinesis and Cytokinesis

υ **karyokinesis** the mitotic division of the nucleus.

υ The mechanical agent of this division is the mitotic spindle, with its microtubules composed of tubulin (the same type of protein that makes up the sperm flagellum).

υ The second process is **cytokinesis** the division of the cell.

υ The mechanical agent of cytokinesis is a contractile ring of microfilaments made of actin (the same type of protein that extends the egg microvilli and the sperm acrosomal process).

υ The mitotic spindle and contractile ring are perpendicular to each other, and the spindle is internal to the contractile ring. The contractile ring creates a cleavage furrow, which eventually bisects the plane of mitosis, thereby creating two genetically equivalent blastomeres.

What is the idea of sperm by Leeuwenhoek?

A.V Leeuwenhoek, the Dutch microscopist who co-discovered sperm in 1678, first believed them to be parasitic animals living within the semen (hence the term spermatozoa, meaning "sperm animals").

Leeuwenhoek (1685) wrote that sperm were seeds (both sperm and semen mean "seed") and that the female merely provided the nutrient soil in which the seeds were planted.

Fluidity of sperm plasma membrane during capacitance?

The fluidity of the sperm plasma membrane is altered by the removal of cholesterol by albumin proteins found in the female reproductive tract.

Steps/Events in Fertilization

Although the details of fertilization vary from species to species, conception generally consists of four major events:

➢ Contact and recognition between sperm and egg. In most cases, this ensures that the sperm and egg are of the same species.

➢ Regulation of sperm entry into the egg. Only one sperm can ultimately fertilize the egg.

➢ Fusion of the genetic material of sperm and egg.

➢ Activation of egg metabolism to start development.

Function of placenta:

Placenta is a disc like structure that forms a connection between the embryo and the uterine wall. The placenta acts to provide oxygen and nutrients to the fetus, whilst removing carbon dioxide and other waste products. It metabolizes a number of substances and can release metabolic products into maternal and/or fetal circulations.

Difference between Monospermy and Polyspermy

1. **Monospermy**:
    - Only **one sperm** enters the egg during fertilization.
    - The **haploid sperm nucleus** combines with the **haploid egg nucleus**, forming a **diploid zygote**.
    - It ensures normal embryonic development by providing the correct chromosome number.
2. **Polyspermy**:
    - Multiple sperms enter the egg.
    - This causes a **triploid or abnormal nucleus**, with extra chromosomes, leading to **non-viable embryos** or developmental issues.

- It disrupts normal fertilization and is considered a pathological condition

Four factors involve in movement of gastrulation

The movements of gastrulation include invagination, involution, ingression, delamination, and epiboly

Metamorphosis in frog

Metamorphosis in frogs is a hormonally controlled process, primarily driven by thyroid hormones. This process transforms the aquatic tadpole into a terrestrial adult frog. Key changes include:

- **Limbs**: Hind and forelimbs develop while the tail recedes.
- **Respiration**: Gills disappear, and lungs grow larger.
- **Feeding**: Tadpole teeth vanish, and the mouth/jaw structure adapts.
- **Diet**: The intestine shortens to accommodate a carnivorous diet.
- **Reproduction**: The development of germ cells begins as metamorphosis ends.
- **Egg Division**: The yolk-rich vegetal hemisphere divides slower, while the animal hemisphere divides faster.

Speed: The rate of metamorphosis depends on environmental factors

Write the examples of transcription factors>

The transcription factors Oct4, Sox2, Klf4 and Nanog act as triggers for the induction of somatic cells to pluripotent stem cells.

What is naive?

An uncommitted cell can be described as naive, meaning that it has received no instructions directing it along a particular developmental pathway.

Or

Naive cell refers to an uncommitted or undifferentiated cell that has not yet received signals to develop into a specific cell type. Naive stem cells, such as those found in early embryonic stages, are capable of maintaining global DNA hypomethylation and retain the ability to differentiate into various cell types depending on the signals they receive.

Macro & micro cleavage in sea urchin

In sea urchins, macro and micro cleavages occur during the fourth cleavage:

- **Macromeres**: Formed by unequal equatorial cleavage in the vegetal tier, these are larger cells.
- **Micromeres**: Also produced by the unequal cleavage in the vegetal tier, these are smaller cells at the vegetal pole.

This division contrasts with the animal tier, where equal meridional cleavage produces uniform **mesomeres**.

What causes this cyclic activity of MPF?

The cyclic activity of Mitosis-Promoting Factor (MPF) is regulated by:

- Cyclin B synthesis activates MPF, while its degradation inactivates it.
- CDK binds Cyclin B and is regulated through phosphorylation and dephosphorylation to control mitosis.
- Translation of stored mRNA ensures Cyclin B availability; its absence halts mitosis.
- MPF drives **karyokinesis** (nuclear division) and **cytokinesis** (cytoplasmic division).

- ## Define: Somite, neural tube & tadpole

- **Somite**:
  Blocks of mesodermal tissue on each side of the notochord. They develop into back muscles, spinal vertebrae, and skin layers during embryonic growth.
- **Neural Tube**:
  The structure that forms from folding neural cells in the embryo. It becomes the brain and spinal cord and helps organize nearby tissues into somites.
- **Tadpole:**
  The larval stage of frogs. It has a mouth and anus, develops swimming muscles, and transforms into the adult amphibian shape

Meroblastic cleavage

Most of the egg cell is full of yolk. The cell divisions do not completely divide the egg, so this type of cleavage is called meroblastic. Since only the cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called discoidal.

Fishes, reptiles, and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg.

The eggs of birds and fishes have only one small area of the egg that is free of yolk (telolecithal eggs), and therefore, the cell divisions occur only in this small disc of cytoplasm, giving rise to the discoidal pattern of cleavage

Some MCQS:

The mammalian egg is also surrounded by a layer of cells called the ✅**cumulus**

The vagal (neck) neural crest lies opposite chick somites___ while the sacral neural crest lies posterior to somite 28. ✅ **1-7**

Paracrine factors such as __and FGF7 are important in normal skin development.✅ **TGF**

The human brain consists of over ___neurons associated with over 10^12 glial cells. ✅ **10^11**

Zoology Legends

In humans, the SOX9 gene, which encodes a___ protein, ✅ **DNA-binding**

The cells at the dorsal most portion of the neural tube become the ___cells.✅ **neural crest**

Identical twins occur in roughly **0.25%** of human births. About __of identical twins have two complete and separate chorions ✅ **33%**

Hox genes can alter the body___✅ **axis**

Osteoblasts, committed bone precursor cells

The gametes and their precursor cells are collectively called **germ cells**, and they are set aside for reproductive function.

Identical twins are of **0.25 %**

Snails have cleavage? **Spiral holoblastic**

Common cleavage in annelids and mollusks are? **Spiral or lophotrocozoa**

An uncommitted cell can be described as **naive**, meaning that it has received no instructions directing it along a particular developmental pathway.

Zoology Legends

Muhammad Zaman

03044756496

Remember me in you Precious Prayer

Jazak Allah

ZOOLOGY
LEGENDS

Team Zoology LEGENDS